Detection of Energy Theft and Defective Smart Meters in Smart Grids Using Linear Regression

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***Abstract*—Energy theft and defective smart meters are significant threats to modern power grids, causing substantial financial losses to utility providers and compromising grid reliability. This paper implements and validates two linear regression-based detection algorithms, the Linear Regression-based Energy Theft Detection and Metering (LR-ETDM) and the Categorical Variable-Enhanced LR-ETDM (CVLR-ETDM), originally proposed by Yip et al. (2017). Using synthetically generated smart meter data for 25 consumers over 48 half-hourly time slots, the LR-ETDM achieves 100% detection accuracy on a constant-theft dataset (Dataset 1) but degrades to 56% on a mixed-theft dataset (Dataset 2) containing time-varying anomalies. The CVLR-ETDM, operating over two days of data (96 time slots), recovers accuracy to 96% by splitting the day into on-peak and off-peak periods, running separate regressions on each, and applying a dual-coefficient classification rule. Additionally, an extended multi-period modification is presented that incorporates severity grading and a composite risk score, enabling utility providers to prioritise field interventions with actionable intelligence.**

***Index Terms*—Energy theft detection, smart meters, linear regression, LR-ETDM, CVLR-ETDM, anomaly coefficient, smart grid, non-technical losses, priority report**

I. Introduction

Energy theft, classified as a Non-Technical Loss (NTL), is a major issue in modern power distribution systems. It includes practices such as meter tampering, bypassing meters, incorrect billing, and faulty readings. These problems lead to large financial losses for utility companies and affect the reliability of the power grid. Globally, losses from energy theft are estimated to exceed $96 billion each year, and in some developing countries, they can account for up to 25% of total revenue loss.

With the introduction of Smart Grids (SG) and Advanced Metering Infrastructure (AMI), traditional mechanical meters have been replaced by smart meters that can communicate data in real time. While this improves monitoring and efficiency, it also creates new risks. Smart meters can be targeted by cyberattacks, allowing attackers to alter energy consumption data sent to utility systems.

Traditional methods for detecting energy theft, such as manual inspections and statistical audits, are costly, time-consuming, and often reactive rather than preventive. Therefore, there is a strong need for automated, data-driven, and reliable detection methods that can continuously analyze the data already collected through AMI systems.

- **LR-ETDM**: It uses a Multiple Linear Regression (MLR) model to detect abnormal consumers by calculating anomaly coefficients based on the difference between supplied energy and reported consumption. This method assumes that any anomaly remains constant over the entire observation period.
- **CVLR-ETDM**: An improved version extends this model by adding categorical (dummy) variables that divide the day into on-peak and off-peak periods. This allows the detection of time-varying theft patterns (such as the h3 attack), which cannot be effectively identified by the basic LR-ETDM approach.

The main contributions of this project are: (1) a MATLAB implementation of both algorithms using synthetically generated data; (2) demonstration of the limitation of LR-ETDM on mixed-theft data, where its accuracy drops from 100(3) a significant improvement in performance, with accuracy recovering to 96% using CVLR-ETDM; and (4) an extended version of the model that includes multi-period analysis, fault severity grading, and a composite risk score to help prioritise actions in practical applications.

II. Smart Grid Architecture and Problem Formulation

A. Neighbourhood Area Network (NAN)

In a NAN, electricity flows from a Distribution Station (DS) to homes through feeders. A main smart meter at the DS records the total energy supplied, $c(t)$, at each time step t. Each house also has a smart meter that records its own usage, $p(t, n)$.

Under normal conditions, the total supply should be equal to the sum of all household consumption, with a small extra term k that accounts for minor technical losses (like line and transformer losses):

$$c(t) = \sum_{n=1}^{N} p(t, n) + k \tag{1}$$

B. Mismatch Formulation

If there is theft or a faulty meter, this balance is disturbed. To measure this, we define a mismatch:

$$y(t) = c(t) - \sum_{n=1}^{N} p(t, n) \tag{2}$$

This mismatch helps us understand what is happening:

- $y(t) > 0$: Less energy is reported than supplied $\Rightarrow$ possible theft
- $y(t) < 0$: More energy is reported than supplied $\Rightarrow$ faulty meter
- $y(t) \approx 0$: System is working normally

C. Anomaly Coefficient Model

Each consumer n is assigned an anomaly coefficient a_n, which shows how much their reported usage differs from actual usage:

$$p_{\text{reported}}(t, n) = \frac{1}{1 + a_n}\, p_{\text{true}}(t, n) \tag{3}$$

The factor $m_n = 1/(1 + a_n)$ tells us what fraction of the true consumption is being reported:

- $a_n = 0$ ($m_n = 1$): Normal usage, no issue
- $a_n > 0$ ($m_n < 1$): Under-reporting $\Rightarrow$ possible theft
- $a_n < 0$ ($m_n > 1$): Over-reporting $\Rightarrow$ faulty meter

III. Methodology

A. LR-ETDM: Linear Regression-Based Detection

1) Model Setup: The problem can be written in a simple linear form:

$$\mathbf{P}\,\mathbf{a} = \mathbf{y} \tag{4}$$

Here, $\mathbf{P}$ contains the meter readings of all consumers over time, $\mathbf{a}$ represents how abnormal each consumer is, and $\mathbf{y}$ is the mismatch between supply and reported usage.

2) Estimation Method: Since $T > N$, the system is solved using a least-squares approach:

$$\hat{\mathbf{a}} = (\mathbf{P}^\top \mathbf{P})^{-1}\,\mathbf{P}^\top \mathbf{y} \tag{5}$$

3) Decision Rule: The decision rule at significance level $\alpha = 0.01$ is:

$$\text{Class}(n) = \begin{cases} \text{THIEF} & p_n < 0.01 \text{ and } \hat{a}_n > 0 \\ \text{FAULTY} & p_n < 0.01 \text{ and } \hat{a}_n < 0 \\ \text{Honest} & p_n \geq 0.01 \end{cases} \tag{6}$$

4) Interpretation: For each flagged consumer, we compute a reporting ratio:

$$m_n = \frac{1}{1 + \hat{a}_n} \tag{7}$$

where, $m_n = 1$: honest, $m_n > 1$: faulty meter, $m_n < 1$: theft

5) Limitation: This method assumes that abnormal behaviour remains the same throughout the day. It works well when theft is constant, but fails when theft happens only during certain periods (e.g., only at night).

B. CVLR-ETDM: Categorical Variable Enhancement

1) Motivation: When consumers steal selectively, e.g. only during off-peak hours, the mismatch $y(t)$ alternates between large positive values (theft period) and near-zero values (honest period). A single constant a_n cannot capture this behaviour, leading to statistical insignificance of true thieves and false detection of normal consumers.

2) Time-Period Decomposition: The day is divided using a binary variable:

- **Off-peak** ($x = 0$): slots 1–16 (00:00–08:00) and 41–48 (20:00–24:00)
- **On-peak** ($x = 1$): slots 17–40 (08:00–20:00)

Each consumer n is assigned two coefficients:

- a_n: base anomaly (valid for the whole day)
- b_n: additional anomaly during on-peak hours

Thus, the effective anomaly during peak hours is $a_n + b_n$.

3) Dual Regression and Regularisation: To model both off-peak and on-peak anomalies simultaneously, the CVLR-ETDM framework constructs a combined regression system. The mismatch equations for the two operating periods are merged into a single matrix formulation:

$$\begin{bmatrix} \mathbf{P}_{\text{off}} & \mathbf{0} \\ \mathbf{P}_{\text{peak}} & \mathbf{P}_{\text{peak}} \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{\text{off}} \\ \mathbf{y}_{\text{peak}} \end{bmatrix} \tag{8}$$

where:

- $\mathbf{P}_{\text{off}}$ represents the smart meter consumption matrix during off-peak periods,
- $\mathbf{P}_{\text{peak}}$ represents the smart meter consumption matrix during on-peak periods,
- $\mathbf{a}$ denotes the baseline anomaly coefficient vector,
- $\mathbf{b}$ denotes the additional peak-period anomaly coefficient vector,
- $\mathbf{y}_{\text{off}}$ and $\mathbf{y}_{\text{peak}}$ are the mismatch vectors for off-peak and peak periods respectively.

The effective anomaly coefficient during off-peak periods is represented by $\mathbf{a}$, while the effective anomaly during peak periods becomes:

$$\mathbf{a}_{\text{peak}} = \mathbf{a} + \mathbf{b} \tag{9}$$

This formulation enables the model to detect both constant and time-varying theft behaviour across different operating periods.

To improve numerical stability and reduce overfitting caused by noisy or highly correlated data, ridge regularisation is incorporated into the regression model. The parameter vector is estimated using:

$$\hat{\boldsymbol{\theta}} = \left(\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I}\right)^{-1} \mathbf{A}^\top \mathbf{y} \tag{10}$$

where:

- $\mathbf{A}$ is the augmented regression matrix,
- λ is the ridge regularisation parameter,
- $\mathbf{I}$ is the identity matrix,
- $\mathbf{y}$ is the stacked mismatch vector,
- $\hat{\boldsymbol{\theta}}$ is the estimated parameter vector containing both $\mathbf{a}$ and $\mathbf{b}$ coefficients.

The regularisation parameter penalises excessively large regression coefficients, thereby stabilising the inversion process and improving robustness against noise and ill-conditioned matrices.

TABLE I
CVLR-ETDM DETECTION SCENARIOS

Sc.	a_n	b_n	$a_n + b_n$	**Classification**
1	≈ 0	≈ 0	≈ 0	Honest
2	$> \delta$	≈ 0	$> \delta$	Thief: all day
3	$< -\delta$	≈ 0	$< -\delta$	Faulty: all day
4	≈ 0	$> \delta$	$> \delta$	Thief: on-peak
5	≈ 0	$< -\delta$	$< -\delta$	Faulty: on-peak
6	$> \delta$	sig.	≈ 0	Thief: off-peak
7	$< -\delta$	sig.	≈ 0	Faulty: off-peak

4) Classification Table: Table I shows how values of a_n and b_n map to different cases. A small tolerance δ is used instead of exact zero to handle noise.

5) Two Days of Data Requirement: With $N = 25$ consumers and $T = 48$ (one day), the system becomes underdetermined. Using two days ($T = 96$) provides sufficient degrees of freedom for reliable estimation and statistical validation.

IV. DATASET DESIGN AND GENERATION

A. Synthetic Data Generation

A synthetic dataset was created to resemble the Irish CER Smart Meter Trial dataset. It includes $N = 25$ consumers (Meter IDs 1001-1025) with half-hourly readings ($T = 48$ per day).

The consumption pattern follows a realistic household trend: low usage during the night, gradually increasing in the morning, peaking in the evening, and then decreasing again at night.

B. Dishonest Consumer Assignments

Out of 25 consumers, 10 (40%) are assigned abnormal behaviour. The assignments vary across datasets, as shown in Table II.

TABLE II
CONSUMER ROLE ASSIGNMENTS PER DATASET

Dataset	**Role**	**Count**	**Meter IDs**
D1 (constant)	h1 Thief	6	1001,1005,1009,1013,1017,1021
D1 (constant)	Faulty	4	1003,1007,1011,1019
D2 (mixed)	h1 Thief	3	1001,1005,1009
D2 (mixed)	h3 Thief	3	1013,1017,1021
D2 (mixed)	Faulty	4	1003,1007,1011,1019

C. Attack Type Definitions

h1: Constant theft: The meter reports only a fixed fraction ($m \in [0.35, 0.60]$) of the actual consumption throughout the day.

h3: Off-peak theft: The meter under-reports only during off-peak hours, while reporting correctly during on-peak hours.

Faulty meter: The meter consistently over-reports consumption, typically by a factor between 1.40 and 1.60.

V. IMPLEMENTATION SUMMARY

A. LR-ETDM

The implementation processes both datasets one by one and evaluates them separately. It builds linear regression models and computes anomaly coefficients directly, without dividing the data into time periods.

B. CVLR-ETDM

The extended implementation uses a modified regression approach that considers different time periods. Compared to the original method, it includes: (i) regularization to improve numerical stability, (ii) flexible threshold to handle noise in the data, and (iii) a modular approach to correctly process multi-day datasets.

VI. RESULTS

A. LR-ETDM on Dataset 1: Constant Theft (h1)

Table III shows the results for Dataset 1. All 25 consumers are correctly classified, giving **100% accuracy**.

The anomaly coefficients for thieves are clearly positive and statistically significant Faulty meters show negative and significant coefficients. All 15 honest consumers have high p-values and multiplier closer to 1, indicating normal behaviour.

TABLE III
LR-ETDM RESULTS: DATASET 1 (CONSTANT H1, ACCURACY = 100%)

ID	$\hat{a}_n$	SE	t	p	m	**Det.**	**Truth**
1001	1.034	0.020	50.69	0.000	0.492	THIEF	THIEF
1002	0.001	0.007	0.07	0.945	1.000	Honest	Honest
1003	-0.337	0.005	-61.80	0.000	1.508	FAULTY	FAULTY
1004	0.011	0.011	1.02	0.316	1.000	Honest	Honest
1005	0.662	0.014	48.64	0.000	0.602	THIEF	THIEF
1006	0.009	0.011	0.83	0.417	1.000	Honest	Honest
1007	-0.381	0.009	-40.77	0.000	1.617	FAULTY	FAULTY
1008	-0.001	0.006	-0.20	0.846	1.000	Honest	Honest
1009	1.507	0.018	84.82	0.000	0.399	THIEF	THIEF
1010	-0.002	0.007	-0.30	0.766	1.000	Honest	Honest
1011	-0.288	0.006	-52.11	0.000	1.404	FAULTY	FAULTY
1012	-0.004	0.006	-0.65	0.523	1.000	Honest	Honest
1013	0.817	0.011	75.62	0.000	0.550	THIEF	THIEF
1014	0.003	0.011	0.27	0.787	1.000	Honest	Honest
1015	-0.008	0.011	-0.69	0.496	1.000	Honest	Honest
1016	0.024	0.012	2.08	0.049	1.000	Honest	Honest
1017	1.225	0.028	43.90	0.000	0.450	THIEF	THIEF
1018	0.011	0.007	1.60	0.124	1.000	Honest	Honest
1019	-0.370	0.006	-64.14	0.000	1.588	FAULTY	FAULTY
1020	-0.003	0.010	-0.35	0.732	1.000	Honest	Honest
1021	1.851	0.025	72.76	0.000	0.351	THIEF	THIEF
1022	-0.012	0.011	-1.06	0.298	1.000	Honest	Honest
1023	-0.004	0.009	-0.50	0.625	1.000	Honest	Honest
1024	-0.001	0.010	-0.09	0.928	1.000	Honest	Honest
1025	0.002	0.007	0.31	0.762	1.000	Honest	Honest
							Accuracy = 100.0% (25/25 correct)

B. LR-ETDM on Dataset 2: Mixed Theft (h1 + h3)

When LR-ETDM is applied to Dataset 2, the accuracy drops significantly to **56% (14/25)**. As shown in Table IV and Fig. 2, the model struggles to correctly identify h3 thieves. Some of these consumers are either missed or wrongly classified. This happens because the mismatch varies across time: it is high during off-peak hours and close to zero during on-peak hours. Since LR-ETDM assumes a constant pattern, it cannot handle this variation, leading to incorrect results.

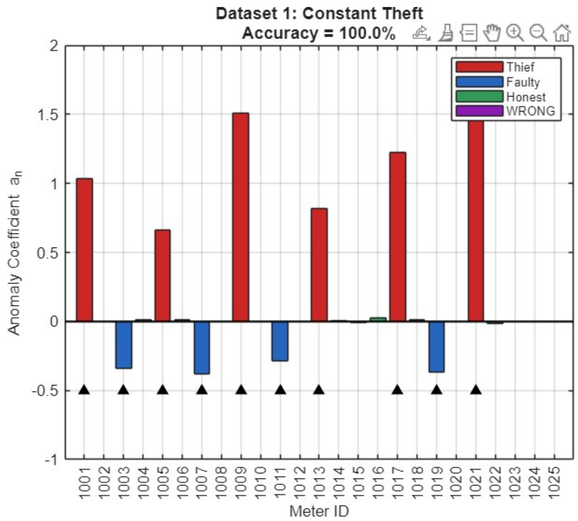


Fig. 1. Anomaly coefficients $\hat{a}_n$ for Dataset 1 (LR-ETDM, 100% accuracy). Red = thieves ($a_n > 0$), blue = faulty meters ($a_n < 0$), green = honest. Black triangles indicate truly dishonest consumers.

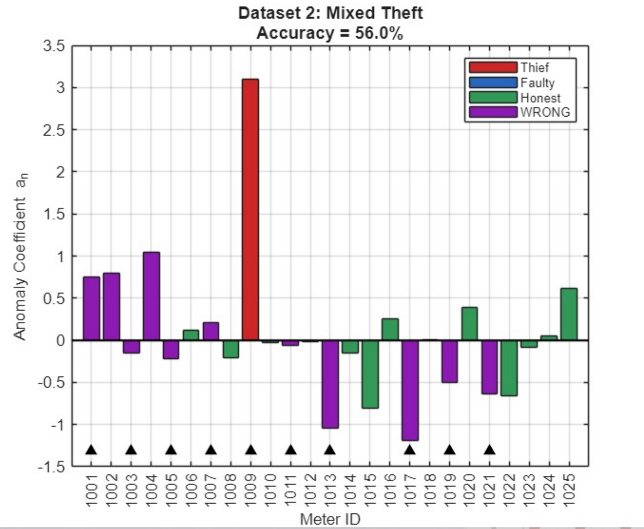


Fig. 2. Anomaly coefficients $\hat{a}_n$ for Dataset 2 (LR-ETDM, 56% accuracy). Purple bars indicate wrong detections. h3 thieves (1013, 1017, 1021) are misclassified, one as FAULTY, two as Honest.

TABLE IV
LR-ETDM RESULTS: DATASET 2 (MIXED H1+H3, ACCURACY = 56%)

ID	$\hat{a}_n$	SE	t	p	m	Det.	Truth
1001	0.750	0.829	0.905	0.375	1.000	**Honest**	THIEF
1002	0.795	0.234	3.403	0.002	0.557	**THIEF**	Honest
1003	-0.156	0.225	-0.693	0.495	1.000	**Honest**	FAULTY
1004	1.043	0.367	2.841	0.009	0.490	**THIEF**	Honest
1005	-0.223	0.531	-0.420	0.679	1.000	**Honest**	THIEF
1006	0.119	0.492	0.241	0.812	1.000	Honest	Honest
1007	0.213	0.359	0.594	0.558	1.000	**Honest**	FAULTY
1008	-0.207	0.271	-0.766	0.451	1.000	Honest	Honest
1009	3.099	0.642	4.831	0.000	0.244	THIEF	THIEF
1010	-0.032	0.258	-0.126	0.901	1.000	Honest	Honest
1011	-0.063	0.227	-0.280	0.782	1.000	**Honest**	FAULTY
1012	-0.012	0.262	-0.044	0.965	1.000	Honest	Honest
1013	-1.041	0.175	-5.943	0.000	-24.6	**FAULTY**	THIEF
1014	-0.155	0.465	-0.333	0.742	1.000	Honest	Honest
1015	-0.801	0.468	-1.714	0.100	1.000	Honest	Honest
1016	0.253	0.480	0.527	0.604	1.000	Honest	Honest
1017	-1.190	0.541	-2.201	0.038	1.000	**Honest**	THIEF
1018	0.006	0.291	0.020	0.984	1.000	Honest	Honest
1019	-0.505	0.243	-2.074	0.050	1.000	**Honest**	FAULTY
1020	0.386	0.400	0.964	0.345	1.000	Honest	Honest
1021	-0.638	0.353	-1.808	0.084	1.000	**Honest**	THIEF
1022	-0.655	0.435	-1.505	0.146	1.000	Honest	Honest
1023	-0.083	0.348	-0.239	0.813	1.000	Honest	Honest
1024	0.052	0.374	0.139	0.891	1.000	Honest	Honest
1025	0.612	0.245	2.497	0.020	1.000	Honest	Honest
Accuracy = 56.0% (14/25 correct), wrong rows in bold							

TABLE V
CVLR-ETDM RESULTS: DATASET 2 (ACCURACY = 96%)

MeterID	Predicted	True GT	Match
1001	THIEF	THIEF	✓
1002	HONEST	HONEST	✓
1003	FAULTY	FAULTY	✓
1004	HONEST	HONEST	✓
1005	THIEF	THIEF	✓
1006	HONEST	HONEST	✓
1007	FAULTY	FAULTY	✓
1008	HONEST	HONEST	✓
1009	THIEF	THIEF	✓
1010	HONEST	HONEST	✓
1011	FAULTY	FAULTY	✓
1012	THIEF	HONEST	× (GT=HONEST)
1013	THIEF	THIEF	✓
1014	HONEST	HONEST	✓
1015	HONEST	HONEST	✓
1016	HONEST	HONEST	✓
1017	THIEF	THIEF	✓
1018	HONEST	HONEST	✓
1019	FAULTY	FAULTY	✓
1020	HONEST	HONEST	✓
1021	THIEF	THIEF	✓
1022	HONEST	HONEST	✓
1023	HONEST	HONEST	✓
1024	HONEST	HONEST	✓
1025	HONEST	HONEST	✓
Accuracy vs GT: 24/25 = 96.0%			

C. CVLR-ETDM on Dataset 2: Recovery

Table V shows the results of CVLR-ETDM on Dataset 2 using two days of data ($T = 96$). The accuracy improves significantly to **96% (24/25)**.

majority of h1 thieves, h3 (off-peak) thieves, and faulty meters are correctly detected, only one is incorrectly classified. This error occurs because its value is very close to the threshold, and small estimation noise causes it to be flagged.

VII. EXTENDED MODIFICATIONS

A. Modification 1: Multi-Class Detection with Severity Grading

The original binary output (flagged / not flagged) is extended into a more informative multi-class framework. Each smart meter is assigned a label from the set:

$$\{\text{HONEST}, \text{THIEF}, \text{FAULTY}, \text{INSPECT}\}$$

where:

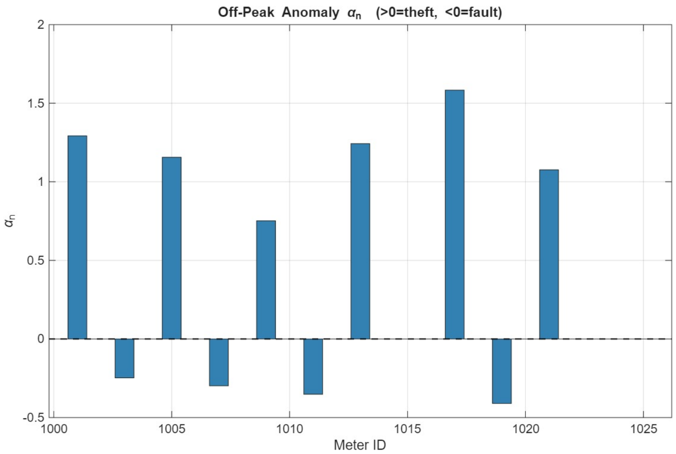


Fig. 3. Off-peak anomaly coefficients (a_n) estimated by CVLR-ETDM for Dataset 2.

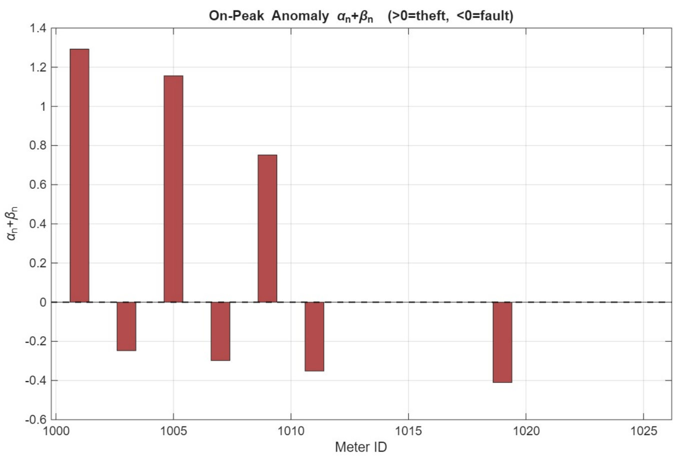


Fig. 4. On-peak effective anomaly coefficients (a_n+b_n) estimated by CVLR-ETDM for Dataset 2.

- **HONEST** indicates normal consumer behaviour,
- **THIEF** represents under-reporting of energy consumption,
- **FAULTY** represents over-reporting caused by meter malfunction,
- **INSPECT** represents ambiguous or mixed anomaly behaviour requiring manual verification.

Instead of using a fixed threshold, the proposed model computes an adaptive anomaly tolerance based on system noise and average consumption:

$$\delta = \frac{\sigma_y}{\bar{P}\sqrt{N}} \tag{11}$$

where:

- σ_y is the robust estimate of mismatch noise,
- $\bar{P}$ is the average smart meter consumption,
- N is the total number of consumers.

The mismatch noise is estimated using the Median Absolute Deviation (MAD):

$$\sigma_y = \frac{\text{median}\left(|y - \text{median}(y)|\right)}{0.6745} \tag{12}$$

This adaptive threshold allows the detector to automatically adjust to different operating conditions and noise levels.

For abnormal meters, a severity level is additionally assigned according to the magnitude of the anomaly coefficient:

$$\text{Severity}(n) = \begin{cases} \text{MILD} & |\hat{a}_n| \in (0, \delta) \\ \text{MODERATE} & |\hat{a}_n| \in [\delta, 2\delta) \\ \text{SEVERE} & |\hat{a}_n| \geq 2\delta \end{cases} \tag{13}$$

This enables prioritisation of suspicious consumers instead of treating all detected anomalies equally.

B. Modification 2: Multi-Period Time Analysis

The original CVLR-ETDM framework separates the day into only two periods (peak and off-peak). In the proposed extension, the model is generalised into a multi-period formulation to better capture time-varying theft behaviour.

Using two days of smart meter data, each day is divided into three operating periods:

- P1: 00:00–08:00
- P2: 08:00–16:00
- P3: 16:00–24:00

A separate anomaly contribution is estimated for each time period, allowing the detector to identify when abnormal behaviour occurs.

For a generalised system with K operating periods, the effective anomaly coefficient of consumer n during period k is expressed as:

$$a_{n,k} = \alpha_n + \beta_{n,k} \tag{14}$$

where:

- α_n represents the baseline anomaly coefficient,
- $\beta_{n,k}$ represents the additional anomaly contribution during period k.

The mismatch equation therefore becomes:

$$y(t) = \sum_{n=1}^{N} a_{n,k(t)}\, p_n(t) \tag{15}$$

where $k(t)$ denotes the active operating period corresponding to time slot t.

This multi-period representation enables the detection of:

- constant theft behaviour,
- peak-only theft,
- off-peak theft,
- and complex time-varying attack patterns.

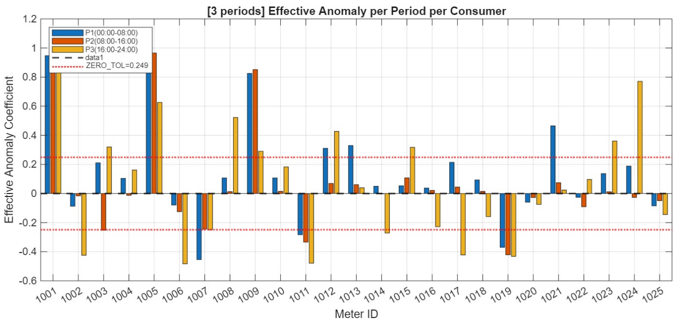


Fig. 5. Anomaly levels of each consumer across three time periods.

C. Composite Risk Score and Priority Report

To improve the practical usability of the proposed detection framework, a composite risk score $R_n \in [0, 10]$ is assigned to every flagged smart meter. Instead of treating all anomalies equally, the risk score combines multiple factors to prioritise suspicious consumers for inspection.

The proposed risk evaluation framework considers:

- Magnitude and severity of the anomaly,
- Consistency of abnormal behaviour across multiple time periods,
- Estimated energy loss caused by the anomaly.

The estimated daily energy loss associated with consumer n is computed as:

$$L_n = \sum_{k=1}^{K} \bar{P}_n \gamma_k |a_{n,k}| \tag{16}$$

where:

- $\bar{P}_n$ is the average daily consumption of consumer n,
- γ_k is the fraction of the day occupied by period k,
- $a_{n,k}$ is the effective anomaly coefficient during period k,
- K is the total number of operating periods.

The temporal consistency factor is defined as:

$$B_n = 1 + 0.5 \left(\frac{\text{Number of Active Periods}}{K} \right) \tag{17}$$

This factor increases the risk score for consumers exhibiting persistent anomalies across multiple operating periods.

The final composite risk score is computed as:

$$R_n = \min\left(10,\ 2S_n B_n + L_n^*\right) \tag{18}$$

where:

- S_n is the numerical severity score,
- B_n is the temporal consistency factor,
- L_n^* is the normalized energy-loss bonus term.

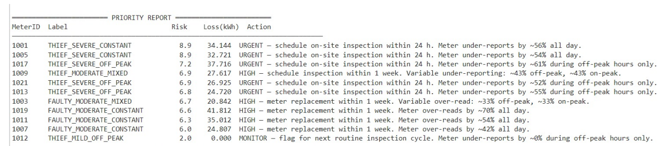
The severity score is assigned as:

$$S_n = \begin{cases} 1, & \text{MILD} \\ 2, & \text{MODERATE} \\ 3, & \text{SEVERE} \end{cases} \tag{19}$$

Based on the computed risk score, inspection priorities are assigned as follows:

- $R_n \geq 7$: **URGENT** (Immediate inspection required)
- $4 \leq R_n < 7$: **HIGH** (Inspection recommended within one week)
- $R_n < 4$: **MONITOR** (Low-priority observation)

This prioritisation strategy enables utility providers to focus resources on consumers causing the highest operational and financial impact.



PRIORITY REPORT

MeterID	Label	Risk	Loss(kWh)	Action
1001	THIEF_SEVERE_CONSTANT	8.9	34.144	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~56% all day.
1005	THIEF_SEVERE_CONSTANT	8.9	32.721	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~54% all day.
1017	THIEF_SEVERE_OFF_PEAK	7.2	37.716	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~61% during off-peak hours only.
1009	THIEF_MODERATE_MIXED	6.9	27.617	HIGH – schedule inspection within 1 week. Variable under-reporting: ~43% off-peak, ~43% on-peak.
1021	THIEF_SEVERE_OFF_PEAK	6.9	26.925	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~52% during off-peak hours only.
1013	THIEF_SEVERE_OFF_PEAK	6.8	24.720	URGENT – schedule on-site inspection within 24 h. Meter under-reports by ~55% during off-peak hours only.
1003	FAULTY_MODERATE_MIXED	6.7	20.842	HIGH – meter replacement within 1 week. Variable over-read: ~33% off-peak, ~33% on-peak.
1019	FAULTY_MODERATE_CONSTANT	6.6	41.812	HIGH – meter replacement within 1 week. Meter over-reads by ~70% all day.
1011	FAULTY_MODERATE_CONSTANT	6.3	35.012	HIGH – meter replacement within 1 week. Meter over-reads by ~54% all day.
1007	FAULTY_MODERATE_CONSTANT	6.0	24.807	HIGH – meter replacement within 1 week. Meter over-reads by ~42% all day.
1012	THIEF_MILD_OFF_PEAK	2.0	0.000	MONITOR – flag for next routine inspection cycle. Meter under-reports by ~0% during off-peak hours only.

Fig. 6. Priority Report showing consumers ranked according to composite risk score and recommended inspection actions.

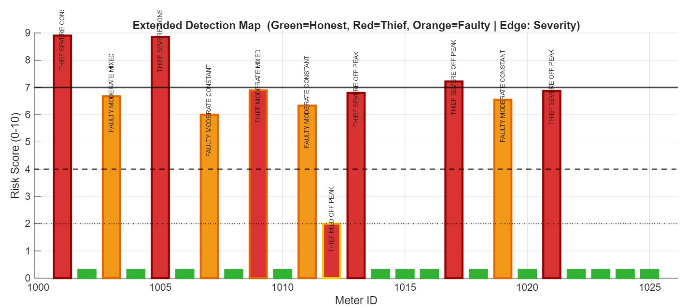


Fig. 7. Extended Detection Map showing risk scores and classifications for all meters.

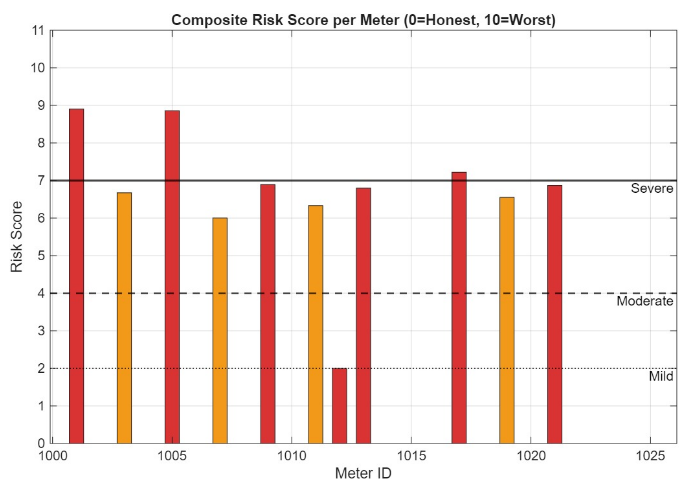


Fig. 8. Risk scores grouped into mild, moderate, and severe levels.

The Extended Detection Map (Fig. 7) visualises the classification, severity level, and risk score of each smart meter. Additionally, Fig. 8 groups the detected anomalies into severity categories for easier operational interpretation and inspection planning.

D. Limitation of Multi-Period Approach

The multi-period model provides improved temporal resolution but introduces additional regression parameters. For a system with N consumers and K operating periods, the approximate number of unknown parameters becomes:

$$N_{\text{params}} = NK \tag{20}$$

However, the total number of observations T remains fixed. Therefore, the effective degrees of freedom reduce to:

$$df = T - NK \tag{21}$$

As the number of operating periods increases, the regression system becomes increasingly underdetermined and statistically unstable. This increases estimation variance and may lead to overfitting, where the model begins fitting noise instead of genuine anomaly patterns.

Consequently, although multi-period analysis provides finer temporal localisation of theft behaviour, it may reduce overall detection accuracy when insufficient data samples are available for each operating period.

VIII. Discussion

A. LR-ETDM Performance

The results match the findings of Yip et al. LR-ETDM performs perfectly on constant-theft data because the anomaly remains stable over time, allowing the model to estimate it accurately. This leads to clear separation between normal and abnormal consumers, with no ambiguity in classification.

However, the accuracy drops to 56% on mixed data. This is because the model assumes that behaviour is constant over time, which is not true for h3-type theft. In such cases, the mismatch changes across time periods, and the model is unable to represent this using a single value per consumer. As a result, the estimates become unreliable. Some honest consumers are wrongly flagged, while some actual thieves are missed. In certain cases, the calculated values become unrealistic, which further indicates that the model is not suitable for time-varying behaviour.

B. CVLR-ETDM Performance

CVLR-ETDM improves performance by analyzing different time periods separately. By splitting the data into consistent subsets (such as off-peak and on-peak), it avoids mixing conflicting patterns. This allows the model to correctly identify h3-type theft, which appears only during specific time periods. As a result, all such consumers are successfully detected.

Only one consumer is incorrectly flagged, which is likely due to small variations in the data near the decision threshold. The use of regularization helps improve numerical stability without significantly affecting the results.

TABLE VI
Algorithm Accuracy Comparison

Algorithm	Dataset 1 (h1)	Dataset 2 (h1+h3)
LR-ETDM	100% (25/25)	56% (14/25)
CVLR-ETDM	100% (25/25)	96% (24/25)
Multi-Period (K=3)	N/A	60% (15/25)

C. Comparison Summary

IX. Conclusion

This project successfully implemented and evaluated the LR-ETDM and CVLR-ETDM algorithms for detecting energy theft and faulty smart meters in smart grids. The key findings are:

1) **LR-ETDM achieves 100% accuracy on constant (h1) theft**, confirming the results of Yip et al. All thieves and faulty meters are correctly identified. The multiplier $m_n = 1/(1 + \hat{a}_n)$ provides a clear interpretation of reported vs actual consumption.
2) **LR-ETDM drops to 56% accuracy on mixed (h1+h3) theft.** This is because the model assumes constant behaviour, which is not valid for time-varying theft patterns. As a result, the method produces incorrect classifications.
3) **CVLR-ETDM improves accuracy to 96%** by analysing different time periods separately. This allows correct detection of both constant and time-varying theft patterns, including previously undetectable cases.
4) **The extended modifications** (multi-period analysis and severity grading) make the results more practical by assigning risk scores and prioritising actions, helping utilities focus on the most critical cases.

Future work can include testing on real-world datasets, integrating real-time detection systems, analysing the impact of measurement noise, and scaling the approach to larger networks.

References

[1] S.-C. Yip, K. Wong, W.-P. Hew, M.-T. Gan, R. C.-W. Phan, and S.-W. Tan, "Detection of energy theft and defective smart meters in smart grids using linear regression,"